cheat_sheet

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User

Make an EXTENSIVE list with all concepts and algorythms of the entire course. MBPO, SARSA, GAIL, you name it. General rule of thumb, if it has an acronym, include it. There are multiple paradigms each with numerous iterations, generally grouped in Value estimation(True/False), Policy Gradient(True/False), On/Off Policy, Model-Based. The exam is generally closed book but we are allowed to write a cheat sheet and even though I went through the entire course I will definetely not remeber all the acronyms or how TD3 differentiates from SAC... So please make sure the list is extensive and includes everything that might be important and points to how a given method stands out/ how it improves on previous methods; also include the formula a given algorythm builds upon.

Gemini

Preparing a cheat sheet for a closed-book exam requires a dense, highly structured breakdown of the material. Deep Reinforcement Learning is notorious for its overlapping acronyms and subtle algorithmic differences.

Here is an extensive, organized compilation of the algorithms and core concepts from your course material. It is grouped by paradigm, highlighting the primary differentiating factors and foundational formulas for each method.

1. Foundations & Tabular Exploration

Markov Decision Process (MDP)

- **Paradigm:** Environment Formalism.
- **Key Concept:** Defines the RL environment as a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \gamma \rangle$.
- **Formula:** Discounted Return: $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$. Bellman Expectation: $v_\pi(s) = \sum_a \pi(a|s)[r(s,a) + \gamma \sum_{s'} p(s'|s,a) v_\pi(s')]$.

Upper Confidence Bound (UCB)

- **Paradigm:** Value-Based / Exploration Strategy.
- **Key Concept:** Replaces naive ϵ-greedy exploration by prioritizing actions with high uncertainty. It uses optimism in the face of uncertainty.
- **Formula:** $A_t = \arg\max_a [Q_t(a) + c\sqrt{\ln t / N_t(a)}]$.

Thompson Sampling

- **Paradigm:** Probabilistic Exploration.
- **Key Concept:** Bayesian approach that maintains a posterior distribution over values. It handles uncertainty more naturally than UCB by directly sampling hypotheses.

2. Tabular Value-Based Methods

Temporal-Difference Learning (TD(0))

- **Paradigm:** Model-Free, Value-Based.
- **Key Concept:** Learns online, step-by-step by bootstrapping off its own estimates. It trades unbiasedness (like Monte Carlo) for lower variance.
- **Formula:** $V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$.

SARSA (State-Action-Reward-State-Action)

- **Paradigm:** Model-Free, Value-Based, On-Policy.
- **Key Concept:** Evaluates the specific behavior policy the agent is currently following, incorporating its exploration noise,.
- **Formula:** $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$.

Q-Learning

- **Paradigm:** Model-Free, Value-Based, Off-Policy.
- **Key Concept:** Learns the optimal greedy policy regardless of what the agent actually does. It suffers from maximization bias.
- **Formula:** $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a') - Q(S_t, A_t)]$.

Double Q-Learning

- **Paradigm:** Model-Free, Value-Based, Off-Policy.
- **Key Concept:** Fixes Q-Learning's maximization bias by maintaining two independent Q-estimates. One selects the action, the other evaluates it.
- **Formula:** $Q^A(S_t, A_t) \leftarrow Q^A(S_t, A_t) + \alpha[R_{t+1} + \gamma Q^B(S_{t+1}, \arg\max_a Q^A(S_{t+1}, a)) - Q^A(S_t, A_t)]$.

3. Deep Value-Based Methods (DQN Family)

Deep Q-Network (DQN)

- **Paradigm:** Model-Free, Value-Based, Off-Policy.
- **Key Concept:** Scales Q-learning with neural networks. Fixes the "Deadly Triad" instability using **Experience Replay** (breaks temporal correlations) and **Fixed Target Networks** (stops moving targets),.
- **Formula (Target):** $y_j = r_j + \gamma \max_{a'} Q_{w^-}(s'_j, a')$.

Double DQN

- **Key Concept:** Adapts Tabular Double Q-Learning for deep networks to fix maximization bias. The online network selects the action; the frozen target network evaluates it.

Prioritized Experience Replay (PER)

- **Key Concept:** Replays surprising transitions more frequently. Sampling probability is proportional to the TD error. Uses importance sampling to correct the introduced bias.

Dueling Networks

- **Key Concept:** Splits the Q-network into two separate heads: one for State Value $V(s)$, one for Action Advantage $A(s, a)$. This allows the network to learn state values even when specific actions aren't taken.

Distributional RL (e.g., C51)

- **Key Concept:** Instead of predicting the expected scalar return, it learns the entire probability distribution of returns. Solves the Distributional Bellman equation.

Noisy Networks

- **Key Concept:** Replaces ϵ-greedy heuristics with learned stochastic linear layers. Shrinks noise where confident, keeps noise where uncertain.

Rainbow

- **Key Concept:** Combines DQN with Double DQN, PER, Dueling, Multi-step returns, Distributional RL, and Noisy Nets into a single agent.

4. Policy Gradients & Actor-Critic (On-Policy)

REINFORCE

- **Paradigm:** Model-Free, Policy-Based, On-Policy (Monte Carlo).
- **Key Concept:** Updates policy parameters by taking the full Monte-Carlo return G_t multiplied by the log-probability gradient. Unbiased but suffers from extremely high variance.
- **Formula:** $\theta \leftarrow \theta + \alpha \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t|S_t)$.

REINFORCE with Baseline

- **Key Concept:** Subtracts a state-value baseline $b(s)$ from the return. It heavily reduces variance without altering the expected gradient.

Actor-Critic (A2C/A3C)

- **Paradigm:** Model-Free, Actor-Critic, On-Policy.
- **Key Concept:** The Critic bootstraps the return using a learned value network, providing a one-step advantage (TD error) as the signal to train the Actor. Lower variance, but introduces bias.
- **Formula (TD Error):** $\delta_t = R_{t+1} + \gamma v_w(S_{t+1}) - v_w(S_t)$.

Generalized Advantage Estimation (GAE)

- **Key Concept:** The policy-gradient equivalent of TD(λ). It exponentially mixes n-step advantage estimates to strictly control the bias-variance trade-off.
- **Formula:** $\hat{A}_t^{GAE(\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$.

Natural Policy Gradient (NPG)

- **Paradigm:** Trust-Region Optimization.
- **Key Concept:** Controls large policy updates by rescaling the gradient using the Fisher Information Matrix. Updates happen in policy space (KL divergence), not just parameter space.

Trust Region Policy Optimization (TRPO)

- **Paradigm:** Trust-Region Optimization.
- **Key Concept:** Enforces a hard mathematical KL divergence constraint to ensure the policy never updates too far at once. Uses heavy second-order math (conjugate gradients).
- **Formula:** $\max \mathbb{E}[r_t(\theta)\hat{A}_t]$ s.t. $D_{KL}(\pi_{\theta_{old}}||\pi_\theta) \leq \epsilon$.

Proximal Policy Optimization (PPO)

- **Paradigm:** First-Order Trust-Region, On-Policy.
- **Key Concept:** The modern workhorse. Simplifies TRPO by removing the second-order math, instead clipping the importance ratio $r_t(\theta)$ to prevent destructive updates. Highly stable and allows multiple training epochs per data batch.
- **Formula:** $L^{CLIP}(\theta) = \mathbb{E}[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$.

5. Continuous Control (Off-Policy Actor-Critic)

Deep Deterministic Policy Gradient (DDPG)

- **Paradigm:** Model-Free, Actor-Critic, Off-Policy.
- **Key Concept:** Essentially "DQN for continuous actions." Learns a deterministic policy $\mu_\theta(s)$ by pushing gradients directly through the critic network. Brittle because the actor easily exploits critic overestimations.

Twin Delayed DDPG (TD3)

- **Paradigm:** Model-Free, Actor-Critic, Off-Policy.
- **Key Concept:** The standard clean baseline for continuous control. It fixes DDPG's brittleness using three tricks:
  1. **Clipped Double-Q:** Takes the minimum of two critics $\min(q_1, q_2)$ to suppress bias.
  2. **Delayed Updates:** Updates the actor less frequently than the critic.
  3. **Target Smoothing:** Adds clipped noise to target actions.

Soft Actor-Critic (SAC)

- **Paradigm:** Model-Free, Actor-Critic, Off-Policy, Max-Entropy.
- **Key Concept:** Adds entropy to the reward objective. The agent is incentivized to maximize reward and act as randomly as possible. Extremely sample-efficient and robust for real-world robotics.

- **Formula:** $J_{ent}(\theta) = \mathbb{E}_{\pi_\theta}[\sum_{t=0}^{T} \gamma^t (R_{t+1} + \alpha H(\pi_\theta(\cdot|S_t)))]$.

6. Core Challenges: Exploration, Credit & State

Exploration Algorithms:

- **RND (Random Network Distillation):** A novelty-based method. Intrinsic reward is the prediction error between a trained network and a fixed, random target network,.
- **ICM (Intrinsic Curiosity Module):** Rewards the agent based on prediction errors of a forward dynamics model in a learned feature space.
- **NGU (Never Give Up):** Combines within-episode novelty tracking and long-term lifelong novelty tracking.

Credit Assignment:

- **RUDDER:** Uses an LSTM sequence model to predict total return and redistributes the reward directly to the causal events that triggered it, bypassing long horizons.
- **HER (Hindsight Experience Replay):** For sparse goal-conditioned tasks. Relabels failed trajectory goals to match the state the agent actually reached, providing a dense learning signal.

7. Imitation Learning & Offline RL

Behavioral Cloning (BC)

- **Paradigm:** Imitation Learning.
- **Key Concept:** Simple supervised learning (predicting expert action from state). Fails easily due to compounding distribution shifts $O(T^2)$,.

DAgger (Dataset Aggregation)

- **Paradigm:** Interactive Imitation Learning.
- **Key Concept:** Rolls out the trained policy, asks the human/expert to label the visited states, and retrains. Forces the training distribution to match deployment.

Conservative Q-Learning (CQL)

- **Paradigm:** Offline RL.
- **Key Concept:** Standard off-policy RL fails on offline datasets because it queries the Q-function on unseen (OOD) actions. CQL fixes this by explicitly pushing down the Q-values for OOD actions and pulling up Q-values for actions in the dataset.

Implicit Q-Learning (IQL)

- **Paradigm:** Offline RL.

- **Key Concept:** Bypasses OOD queries entirely using an asymmetric expectile regression. It implicitly calculates the best value in the data without explicitly evaluating unseen actions.

Decision Transformer

- **Paradigm:** Offline RL / Sequence Modeling.
- **Key Concept:** Abandons Bellman backups. Uses a GPT-style transformer to autoregressively predict actions based on the sequence of (return-to-go, state, action),. Cannot "stitch" separate sub-trajectories together.

8. Inverse RL & Reward Learning

Max-Margin IRL

- **Paradigm:** Inverse RL.
- **Key Concept:** Solves for a reward function that makes the expert's feature expectations score higher than any other policy by a defined margin,.

Maximum-Entropy IRL (MaxEnt IRL)

- **Paradigm:** Inverse RL.
- **Key Concept:** Resolves reward ambiguity probabilistically. Chooses the reward that makes the trajectory distribution as uniform as possible while still matching the expert's feature expectations.

Generative Adversarial Imitation Learning (GAIL)

- **Paradigm:** Adversarial IRL.
- **Key Concept:** Uses a GAN architecture. A discriminator learns to differentiate expert data from agent data, and the agent uses the discriminator's output as the reward signal,.

RLHF (RL from Human Feedback)

- **Paradigm:** Preference-Based / LLM Alignment.
- **Key Concept:** Trains a Reward Model using the Bradley-Terry preference loss. A policy (e.g., an LLM) is then optimized against this reward using PPO, stabilized by a KL-penalty to an initial SFT model to prevent reward hacking,.

Direct Preference Optimization (DPO)

- **Paradigm:** LLM Alignment.
- **Key Concept:** Bypasses PPO and the explicit reward model. Implicitly expresses the optimal reward in terms of the policy, mapping preference pairs directly into a stable supervised loss,.
- **Formula:** $\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(\tau_w,\tau_l)}[\log \sigma(\beta \log \frac{\pi_\theta(\tau_w)}{\pi_{ref}(\tau_w)} - \beta \log \frac{\pi_\theta(\tau_l)}{\pi_{ref}(\tau_l)})]$.

Group Relative Policy Optimization (GRPO)

- **Paradigm:** LLM Alignment.

- **Key Concept:** A PPO variant that completely eliminates the need for an expensive Critic/Value network. It standardizes the rewards across a generated group of outputs to construct the advantage.
- **Formula:** $\hat{A}_i = \frac{r_i - \text{mean}(r_{1:G})}{\text{std}(r_{1:G}) + \delta}$.

9. Model-Based RL (MBRL)

Monte Carlo Tree Search (MCTS)

- **Paradigm:** Model-Based Planning.
- **Key Concept:** Asymmetric lookahead search prioritizing promising subtrees. Built on 4 phases: Selection, Expansion, Simulation, Backpropagation,. Uses UCT as the tree policy.

AlphaZero

- **Paradigm:** Model-Based RL / Self-Play.
- **Key Concept:** Combines MCTS with a single network predicting both policy priors and state values. MCTS acts as the policy improvement operator, and the neural network amortizes this search via self-play.

Model Predictive Control (MPC) with CEM

- **Paradigm:** Model-Based Continuous Planning.
- **Key Concept:** Receding-horizon continuous planning. Generates action sequences using the Cross-Entropy Method (CEM) against a model, executes only the first step, then completely replans.

Dyna / Dyna-Q

- **Paradigm:** Model-Based Learning.
- **Key Concept:** The foundational loop. Uses real experience to update the model and the value function, then runs multiple synthetic updates by generating "imagined" experience from the model,.

Model-Based Policy Optimization (MBPO)

- **Paradigm:** Model-Based Learning.
- **Key Concept:** A modern Dyna framework. It heavily restricts compounding model errors by executing very short imagined rollouts ($k \approx 1 - 5$) initialized exclusively from real states in the replay buffer, then trains SAC on the mix.

PETS (Probabilistic Ensembles with Trajectory Sampling)

- **Paradigm:** Model-Based Planning.
- **Key Concept:** Tackles epistemic uncertainty by combining MPC planning with an ensemble of probabilistic transition models. Trajectories are sampled through the ensemble to safely guide the search.